



1/7/2026

ASSIGNMENT DAY 3

RAM & Storage Fundamentals.

TASK 1.

❖ Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

➤ Task: Comparison between DDR3, DDR4, and DDR5 RAM:

<u>Feature</u>	<u>DDR3</u>	<u>DDR4</u>	<u>DDR5</u>
Speed	800–1600 MHz, slower data transfer speed	1600–3200 MHz, better speed and performance	4800 MHz and above, highest speed and faster data transfer
Voltage	1.5V, consumes more power	1.2V, more power efficient than DDR3	1.1V, most power efficient among the three
Bandwidth	Lower bandwidth	Improved bandwidth	Highest bandwidth for better performance
Typical Use	Older computers and laptops used for basic tasks	Modern PCs used for gaming, multitasking, and daily work	Latest high-performance PCs, gaming systems, AI, and servers

Technical Summary:

- ✓ DDR3, DDR4, and DDR5 represent the improvement of RAM technology over time. DDR3 is an older generation, DDR4 is commonly used in modern computers, and DDR5 provides the best speed, efficiency, and performance for advanced systems.

TASK 2.

- ❖ Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully, Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.

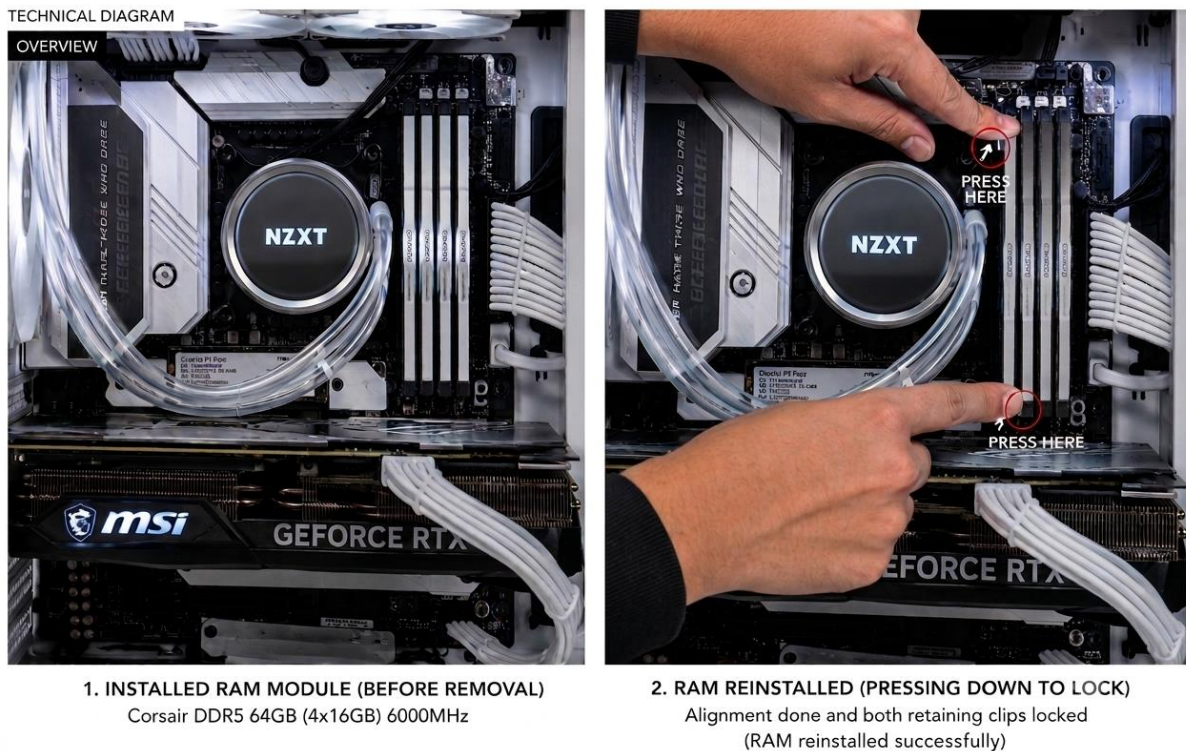


Figure 1: Installed RAM module before reinstallation.

- ✓ **Execution & Verification Report Safety** : The computer was completely turned off, and the power cable was disconnected before opening the system.

- ✓ **Identification & Handling:** The RAM module was located on the motherboard. The retention clips were gently pressed to remove the RAM module carefully.
- ✓ **Reinstallation:** The RAM module was aligned correctly with the slot and pressed firmly until the retention clips locked it into place.
- ✓ **System Verification:** The power cable was connected again, and the computer was turned on.
- ✓ **Result:** The system booted successfully without any memory errors, confirming that the RAM module was installed correctly.

TASK 3.

❖ **List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.**

➤ **Task: Key Differences Between HDD, SSD, and NVMe Storage:**

<u>Feature</u>	<u>HDD (Hard Disk Drive)</u>	<u>SSD (Solid State Drive)</u>	<u>NVMe SSD (Non Volatile Memory Express)</u>
Speed	Slowest. It takes more time to boot the system and open files.	Much faster than an HDD. The computer starts quickly and apps open faster.	Fastest storage type. It provides very quick boot times, file transfers, and app loading.

Cost	Cheapest and available in large storage capacities.	Costs more than an HDD but offers much better performance.	Most expensive because it delivers the highest speed and performance.
Best Use Case / App	Best for storing photos, videos, movies, music, and backup files.	Best for daily apps like Microsoft Office, Google Chrome, and other everyday tasks.	Best for high-performance apps like Adobe Premiere Pro and games such as FIFA or PUBG PC, where fast loading is important.

Technical Summary:

- ✓ HDD is a good choice when you need a lot of storage at a low cost. SSD is the best option for everyday use because it makes the computer faster and more responsive. NVMe SSD offers the highest performance, making it ideal for gaming, video editing, and other tasks that require very high speed.

TASK 4.

❖ **Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly, Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.**

➤ **Storage Drive Removal and Reinstallation :**



Figure : Step by step process of removing, identifying, reinstalling, and verifying the storage drive.

Objective :

- ✓ To remove the storage drive, identify its connector type, reinstall it correctly, and confirm that the system detects the drive after reassembly

Hardware Specifications :

- **Motherboard** : ASUS ROG Maximus Z790 Formula
- **Storage Drive** : Samsung 990 Pro 2TB M.2 NVMe SSD

Step by Step Procedure

- ✓ **Drive Removal (Panel 1)** : The computer was turned off and disconnected from the power supply. The M.2 heatsink was removed carefully to access the storage drive.
- ✓ **Storage Drive Identification (Panel 2)** : The SSD was removed carefully from the motherboard. The connector type was identified as M.2 NVMe SSD.

- ✓ **Reinstallation (Panel 3)** :The SSD was aligned with the M.2 slot and inserted correctly. It was secured in place, and the heatsink was reinstalled.

- ✓ **System Verification (Panel 4)** : After reassembling the computer, the system was turned on. Windows successfully detected the SSD in Disk Management, confirming that the storage drive was installed correctly.

Result:

The storage drive was reinstalled successfully, and the system recognized it without any issues.